SUGGESRED SOLUTIONS TO 2019 PHYSICS H2P3 PRELIM EXAMS

Q1(a)
Base quantities are physical quantities on the basis of which other quantities can be expressed, e.g. velocity can be expressed in terms of the base quantities of length and time.

Any one of the 7 base quantities: Length, mass, time, electric current, temperature, luminous intensity, amount of substance.

1(b)
$$\frac{\Delta C}{C}\text{x}100 = \left[\frac{\Delta\left(\frac{V}{t}\right)}{\left(\frac{V}{t}\right)} + 4\frac{\Delta r}{r} + \frac{\Delta L}{L} + \frac{\Delta \rho}{\rho}\right] \text{ x } 100$$
$= 3\% \ + \ 4(1\%) + 0.5\% + 2\ \%$
The radius of the pipe has the most significant uncertainty which accounts for the greatest percentage uncertainty of 4 % for C.

1(c)(i)
$F = ma$
$a = 0.20/2.0 =$ **0.10 m s^{-2}**
in a **westerly** direction

1(c)(ii)
Resolving the initial velocity in the easterly direction,
$u_e = 1.41 \cos 45° = 1.0 \text{ m s}^{-1}$
Resolving the initial velocity in the northerly direction,
$u_n = 1.41 \cos 45° = 1.0 \text{ m s}^{-1}$

At the end of 10 s, taking east as positive direction,
final velocity in the easterly direction,
$v_e = u_e + at$
$= 1.0 + (-\ 0.10)(10) =$ **0 m s^{-1}**
Therefore, the resultant velocity = **1.0 m s^{-1}**
in a **northerly** direction.
Q2(a)